

CHICAGO JOLIET, IL, USA

WMO: 725345

Lat: 41.518N

Lon: 88.176W

Elev: 177

StdP: 99.21

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 14834

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.1	-23.9	0.4	-17.9	-21.1	0.6	-15.0	9.3	-0.1	8.5	-0.2	2.4	280	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.0	24.2	32.0	23.7	30.5	22.8	26.2	31.0	25.1	29.8	24.1	28.3	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.5	29.0	23.7	18.9	27.7	22.7	17.8	26.8	82.5	31.0	77.2	29.9	73.2	28.1	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.7	7.7	6.9		-23.9	35.0	4.4	1.8	-27.1	36.2	-29.6	37.3	-32.1	38.3	-35.2	39.5	
				-24.2	27.9	4.1	1.1	-27.1	28.7	-29.5	29.3	-31.9	29.9	-34.8	30.7	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-4.0	-2.8	3.9	10.3	16.0	21.3	23.4	22.4	18.8	11.6	5.2	-0.8
	DBStd	10.90	6.76	6.61	6.23	5.29	5.07	3.69	3.20	3.01	4.31	5.13	5.21	5.82
	HDD10.0	1609	436	359	209	60	7	0	0	0	1	42	161	336
	HDD18.3	3358	693	591	450	246	108	13	2	4	46	218	395	592
	CDD10.0	1791	1	2	19	68	193	339	414	383	263	91	16	2
	CDD18.3	497	0	0	2	4	36	102	157	128	58	9	0	0
	CDH23.3	4704	0	0	13	69	366	983	1560	1107	526	79	1	0
	CDH26.7	1588	0	0	1	12	111	343	582	359	163	17	0	0
Wind	WSAvg	2.8	3.2	3.2	3.4	3.7	3.0	2.4	2.0	1.8	2.1	2.6	3.0	3.1
Precipitation	PrecAvg	910	46	41	56	91	106	107	98	100	78	74	65	48
	PrecMax	1153	103	143	128	176	187	251	196	243	320	201	149	125
	PrecMin	602	11	9	17	37	19	23	30	41	15	13	10	13
	PrecStd	126	25	27	28	38	41	55	39	41	60	48	39	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	17.1	25.5	27.9	32.2	34.2	35.9	34.0	33.5	29.0	21.3	17.2
		MCWB	11.9	12.1	17.1	17.7	22.0	23.7	25.1	24.9	23.7	21.6	14.2	14.8
	2%	DB	10.9	12.6	20.3	25.3	29.9	32.4	33.0	32.2	31.1	25.7	17.9	12.6
		MCWB	9.0	9.4	14.3	15.9	20.2	23.2	25.0	24.2	22.7	18.1	13.3	10.6
	5%	DB	7.5	8.8	16.7	22.8	27.4	30.9	31.8	30.7	28.2	22.8	16.1	9.8
		MCWB	6.1	6.3	12.2	14.7	19.0	22.3	24.5	23.3	21.0	17.1	12.3	8.1
	10%	DB	4.9	5.8	13.6	20.3	25.1	28.3	30.1	28.4	26.4	20.7	13.3	7.3
		MCWB	3.3	3.6	9.7	13.8	18.5	21.4	23.2	22.0	19.7	16.1	9.8	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	14.2	17.8	20.1	24.1	26.7	28.0	27.5	25.5	22.9	17.1	15.0
		MCDB	14.3	15.4	23.1	25.7	29.5	31.8	32.2	32.1	30.7	26.9	19.8	16.4
	2%	WB	8.8	10.0	15.6	17.9	22.1	24.9	26.7	25.7	24.2	20.8	14.7	11.1
		MCDB	10.3	12.5	20.1	23.0	26.8	29.8	31.3	30.1	28.8	22.8	16.9	12.2
	5%	WB	5.7	6.7	13.3	16.1	20.8	23.6	25.3	24.4	22.7	18.6	12.5	8.0
		MCDB	7.6	8.9	16.3	21.3	25.0	28.2	29.9	28.5	26.4	21.4	15.1	8.9
	10%	WB	3.2	3.7	10.2	14.3	19.6	22.6	24.3	23.4	21.5	16.6	10.6	5.5
		MCDB	4.7	5.6	13.0	19.7	23.7	27.0	28.5	27.1	25.0	19.8	13.2	7.2
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.9	9.8	11.6	11.9	11.5	11.1	10.9	11.9	10.9	9.3	7.3
		MCDBR	10.4	12.5	13.7	15.6	14.4	13.3	12.3	12.6	13.5	14.3	12.9	10.0
		MCWBR	8.8	9.7	9.3	8.9	7.2	6.2	6.0	6.3	6.3	8.3	9.1	8.7
	5% WB	MCDBR	9.9	11.5	12.4	13.8	11.8	11.2	11.0	10.5	11.2	11.1	11.1	9.4
		MCWBR	8.6	9.6	9.2	9.1	7.4	6.3	6.5	6.0	6.0	8.2	9.0	8.7
Clear-Sky Solar Irradiance	taub	0.295	0.315	0.349	0.403	0.433	0.439	0.454	0.445	0.416	0.359	0.314	0.296	
	taud	2.424	2.372	2.373	2.233	2.203	2.230	2.184	2.201	2.252	2.412	2.488	2.447	
	Ebn at Noon	859	892	898	867	846	840	822	817	815	830	830	826	
	Edh at Noon	78	96	109	134	142	139	144	137	122	92	74	71	
All-Sky Solar Radiation	RadAvg	1.65	2.53	3.59	4.65	5.30	6.05	6.05	5.32	4.46	2.95	1.90	1.40	
	RadStd	0.17	0.34	0.27	0.38	0.44	0.43	0.33	0.30	0.38	0.33	0.18	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page